# Nizar Esmat Fawzy

Software Engineer
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## PROFESSIONAL SUMMARY


- Software Engineer specializing in scalable backend services, production REST APIs, and distributed architectures using Python (Django REST Framework), NestJS, and PostgreSQL.
- Experienced in end-to-end system ownership: relational data modeling, real-time events, third-party integrations, and automated testing pipelines.
- Strong engineering foundation in data structures, algorithms, and database query optimization, with a focus on writing reliable, low-latency production code.


## PROFESSIONAL EXPERIENCE

**Opream** | *Software Engineer* | Cairo, Egypt **May 2025 - Present**

- Design, build, and maintain production backend services in Python/Django REST Framework and NestJS on PostgreSQL, collaborating in an agile engineering team.
- **Mostaqbel Watan (Social Platform):** Built a private community platform featuring token authentication with role-based permissions, real-time Socket.io chat, self-hosted Jitsi video calls, and automated Facebook Graph API data ingestion.
- **OnWay (Ride-Hailing):** Implemented ride lifecycle state machines and integrated the Fawry payment gateway, handling asynchronous webhook callbacks, transaction verification, and failure retries.
- **Kalia (Fintech & AI):** Built backend services and data scraping pipelines to aggregate banking promotions and credit card offers, integrating an AI agent that analyzes user spending to recommend tailored financial deals.
- **Al-Azhar Education Platform:** Developed lesson management APIs, structured relational course models, and implemented granular role-based access control (RBAC) across student and instructor workflows.
- **Engineering Standards:** Profiled Django ORM and SQL queries to eliminate N+1 bottlenecks and optimize indexes; wrote unit/integration tests with `pytest` and `Jest`, and maintained Dockerized development environments.

**iSchool & WE School** | *Software & Web Development Instructor* | Cairo, Egypt **Aug 2023 - Nov 2024**

- Taught computer science fundamentals, algorithmic problem solving in C++, and modern web development (JavaScript, HTML, CSS) to aspiring developers.

## PROJECTS

**ECPC Movement Control** | *NestJS, PostgreSQL, TypeORM, Socket.io, React, JWT* GitHub

- Led the **system design** of a real-time volunteer dispatch platform for ECPC; built an automated scoring algorithm to assign requests by hall proximity with live Socket.io updates.

**Budget Buddy** | *React Native, SQLite* GitHub

- Engineered an offline-first mobile app in React Native; applied SQLite **query optimization** and indexing to compute real-time expense analytics and budget alerts with zero lag.

**Smart Learning Management System** (Graduation Project) | *Python, scikit-learn, pandas, NumPy*

- Designed an academic prediction pipeline using linear regression to forecast student final grades from historical coursework trends, enabling early faculty interventions to improve GPA.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, C++
**Backend Frameworks:** Django, Django REST Framework (DRF), NestJS, Node.js, Express.js, Socket.io
**Databases & Storage:** PostgreSQL, MongoDB, SQLite, Schema Design, Indexing, Query Optimization
**DevOps & Tools:** Docker, Linux/UNIX, Nginx, Gunicorn, Git, GitHub, Postman, CI/CD
**Testing & Architecture:** pytest, Jest, Unit & Integration Testing, RESTful API Design, Microservices
**Frontend & Mobile:** React, React Native, HTML5, CSS3 **Spoken Languages:** Arabic (Native), English (Fluent)

## EDUCATION

**Information Technology Institute (ITI)** | *Full Stack Python* | Smart Village, Egypt **Nov 2024 - Mar 2025**

- Intensive 9-month professional program covering full-stack architecture, Django REST Framework, relational database design, and production engineering.

**Nahda University** | *B.Sc. in Computer Science* | Beni-Suef, Egypt **Sep 2019 - Jul 2023**

## CERTIFICATIONS

Python Professional Track – ITI | SQL & PostgreSQL – Udemy | Node.js – Udemy | MERN Stack – Route Academy

## VOLUNTEERING & ACHIEVEMENTS

- **2nd Place** – Nahda University ECPC Qualifications (2021, 2022).
- **Organizing Volunteer** – ICPC, ECPC, and ACPC Regional Programming Contests (2022 – 2024).
- **Competitive Programming** – Solved 300+ algorithmic problems in C++ on Codeforces.